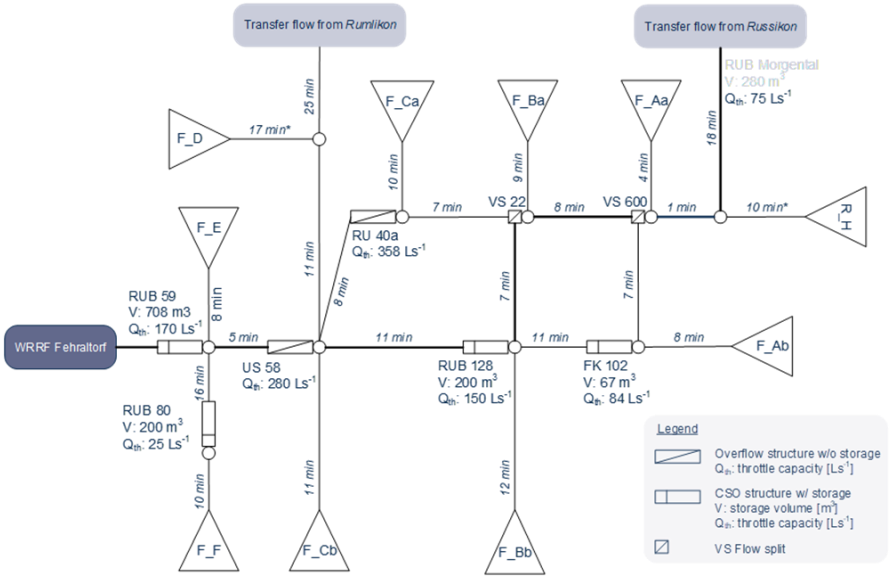
1 Flow scheme and characteristics of the subcatchments

1.1 The drainage system

The main flow direction in Fehraltorf is from north to South East, i.e. from RUB Morgenthal to the WWTP is clearly marked in bold on the flow scheme, passing through VS22, RUB128, and US58 (Figure S 2-1). The flow time for each section can be found in the VGEP 2016 [REF] document, which also includes the numbers used in the flow scheme. Additionally, the flow scheme displays the storage volumes of the special structures and their throttle capacities, providing valuable information for the system's performance.



	Catchment	Fred [hared]	Inhab.
Rumlikon*/Russikon	R_A*	3.6	453
	R_B	5.71	516
	R_C	1.98	275
	R_D	2.32	706
	R_E	1.63	203
	R_F	5.58	1'047
	R_H	-	120
	Summe	20.8	3'320
Fehraltorf	F_Aa	0.78	151
	F_Ab	2	292
	F_Ba	5.9	831
	F_Bb	3.6	803
	F_Ca	5.5	666
	F_Cb	1.58	914
	F_Cd	-	73
	F_D	2.08	881
	F_E	3.67	656
	F_F	7.57	962
	Summe	32.7	6'229
Total		53.5	9'549

Figure S 1-1 Flow topology (left) and aggregated information on the subcatchments (right).

The technical elements of the drainage system include the following special structures: 3 CSO tanks, 1 interceptor sewer, 6 combined sewer overflows and several flow diversion structures. Manual and remotely controllable flow limiters facilities are available. The operation of the basins is ensured manually on site, there is no real-time control scheme in place.

In 2012, emission from the CSOs were estimated with a SIMBA# hydrological model, using rainfall data from the MeteoSwiss rain gauge at Kloten. Although the rain gauge is located at an approximate distance of 14km, the plausibility of the model was high, with ca. 10% difference regarding the number of CSO spills between 2008-2012. Main emissions occurred from CSO tanks Morgental and ARA/WWTP (Figure S 2-2).

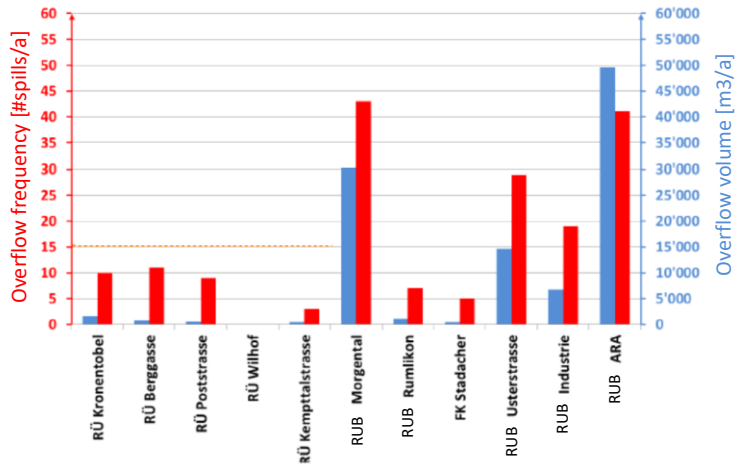


Figure S 1-2 CSO behaviour estimated in the general drainage plan (VGEP 2016).

As receiving water for the discharges from Special structures are waters Fälibach, All the watercourses in the municipality flow together and leave the municipality as Kempt. The influence of the wastewater treatment plant on the watercourses is small in terms of the substance load. The influence of the WWTP on the watercourse is small in terms of the material load, but very large with a design inflow of 200 l/s, which affects the hydraulic load. The monthly average discharge is in the range of 367 l/s. while the minimum discharge(Q347) is 69 l/s.

In the GEP 2014, the discharge coefficients and population densities per building zone used in the old GEP (1999) were adapted to the new situation. According to the local planners the new zoning plan contains an increase in density of 10% and an increase in population density of 15% are expected. The F_hared and population densities were increased accordingly (see Figure S 2-3).

From a hydraulic point of view, the drainage network functions very well in its as-is and fully developed state. There are no hydraulic bottlenecks in the network.

1.2 Engineering assumptions on municipal water use, pollution and infiltration

The general drainage plan reports information on water use, pollution and extraneous flows, which has been obtained by analysing the data the WWTP for the years 2008 - 2012:

- Mean dry weather inflow: 38 l/s, mean value of all dry weather days.
- Mean infiltration ratio: 42 %.
- Mean COD load: 1454 kg COD/d , mean value for the years 2008 - 2012.
- Mean NH4 load: 64.8 kgN/d , mean value for the years 2008 - 2012.
- Mean P"" load: 19.1 kgP/d , mean value for the years 2008 - 2012.

On this basis, the water consumption per inhabitant and the degree of pollution of the foul sewage (FS) of the wastewater was calibrated.

- Water consumption 184 l/(EW*d)
- COD concentration FS 696 mg/l
- NH4concentration FS 31.8 mg/l
- Ptot concentration FS 8.8 mg/l

The maximum dry weather flow was measured to 90 l/s. The hydraulic capacity of the WWTP was assumed to be 200 l/s.

2 Retention basins and flow control structures in the Fehraltorf urban drainage system

2.1 Rüb Morgenthal

RUB Morgental plays a central role in the urban drainage system of Russikon. The structure's throttle flow setting (75 Ls^{-1}) strictly constrains the carry-on flow towards the Fehraltorf system. The detention tank has a total volume of 280 m^3 (tank and inflow), the flow limiter (Q_{th}), i.e. carry-on flow to the WWTP, is fixed through a vortex-type flow limiter at 75 Ls^{-1}. The spillway (nominal pipe size 1000 mm) discharges into the Rohrbach creek.

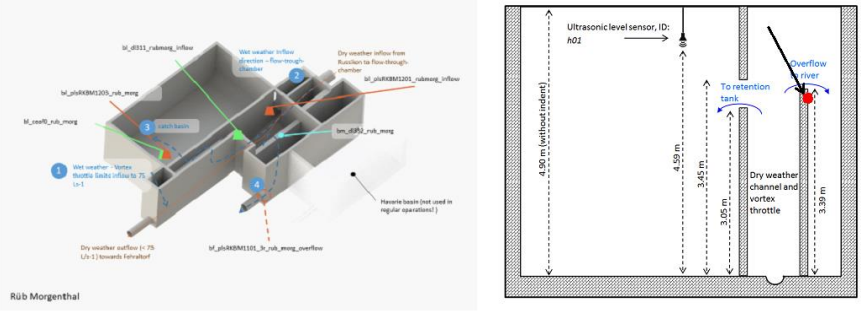


Figure S 2-1 left: 3D drawing of RUB Morgental, incl. Sensor positioning and major flow path during wet weather. right: Cross-section through the overflow structure RUB Morgental (not drawn to scale) including inflow chamber, detention tank, overflow to receiving water and installed monitors

If the inflow exceeds Q_{th}, wastewater backs up within the flow-through chamber(1 & 2) until it reaches a water level of 3.05 m. If the water level exceeds 3.05 m, excess water overflows into the detention tank (catch basin ; 3) until it reaches a fill level of about 3.39 m. At this point, an overflow takes place(4) . The catch basin is emptied by gravity, i.e. through a valve that opens as soon as the throttle discharge rate has decreased to nearly dry weather conditions, i.e. below a threshold of 30 Ls^{-1}. The retention tank is cleaned on demand through a flushing bucket system.

The rainwater overflow basin also uses the clarification basin built in 1963 as an emergency basin, which has an additional storage volume of 145 m3 . However, the emergency basin is not decisive for regular operation.

2.2 FK102 Stadacher

The in-sewer storage with overflow at the upstream end(FK 102 – Stadacher) runs parallel to the Luppmen for about 250m and has a storage volume from 67 m^3. The discharge of the in-sewer-storage channel is throttled at the end by a vortex throttle to 35-40 l/s, which flows towards the wastewater treatment plant. A rake additionally filters solids before the overflow into the watercourse on the overflow edge with a length of 6m. The storage channel is emptied automatically after the inflow has been reduced.

The relevance of the structure in the overall hydraulic context is low. Due to the high overflow edge at the distribution structure 600 (VS600) and the additional vortex throttle at Rüb Morgenthal (limited to 75 ls^{-1}) beforehand, an overflow only occurs during very extreme events.

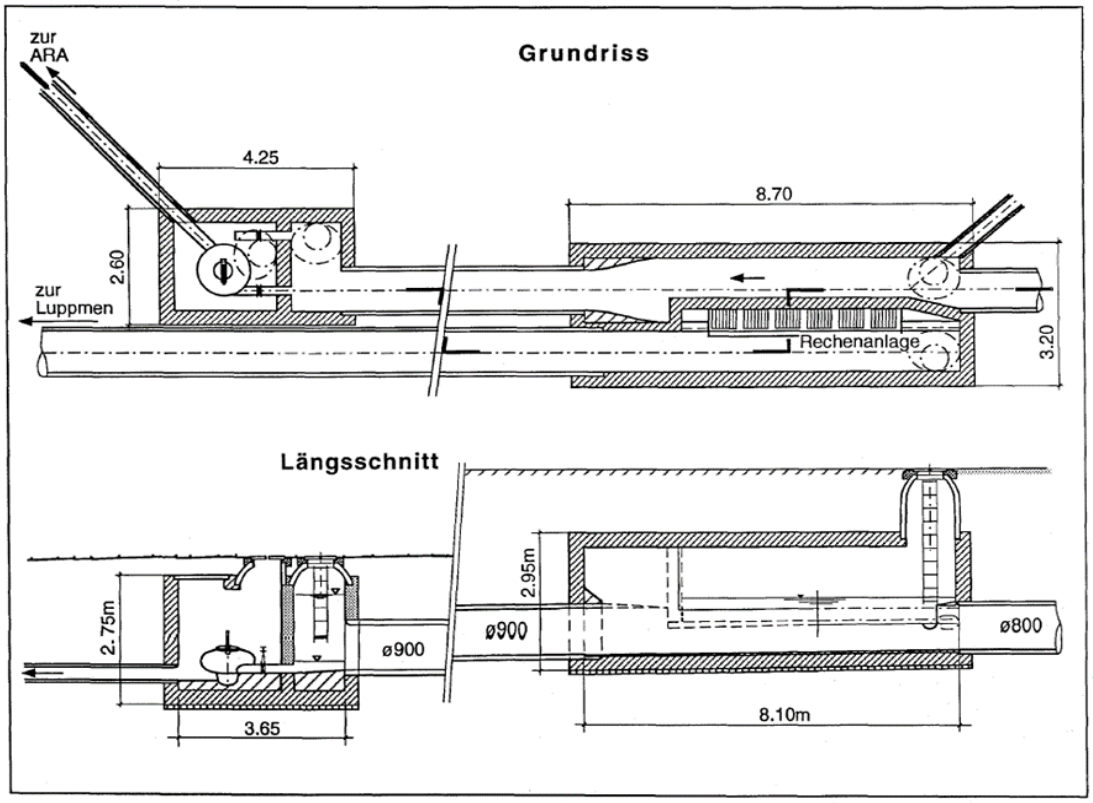
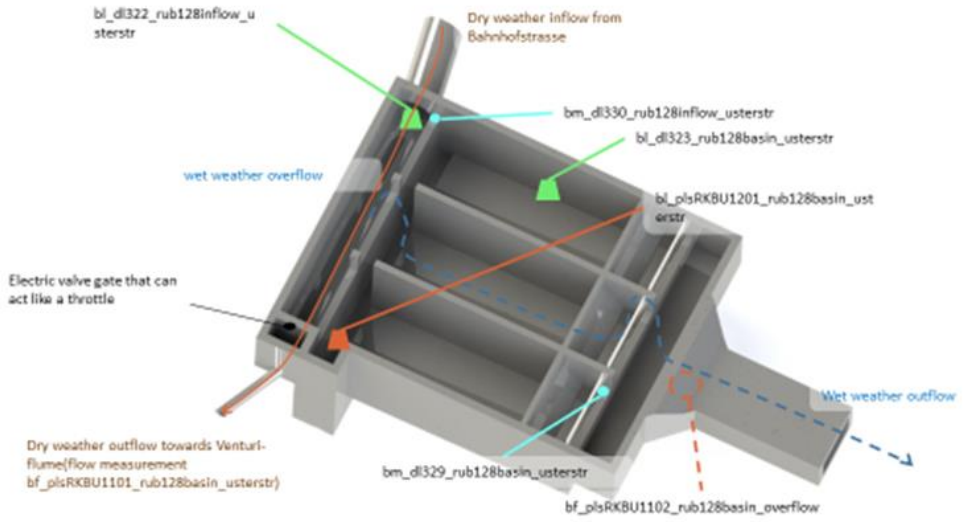


Figure S 2-2: top: planar view onto the storage sewer, bottom: longitudinal cross-section from REF_Krejci (1994)

2.3 Rüb 128 Usterstrasse

The flow-through basin built in 1987 in the tributary has a storage volume of 200 m3. It allows approx. 150 l/s dry weather flow unhindered by the throttling (nominal width reduction from 1250 mm to 400 mm); in addition, the flow can be further limited by means of electric gate valves. Further in the direction of the sewage treatment plant, there is a Venturi flume for flow measurement. In the event of rain, the basin fills up and discharges via 3 overflow edges, each 3 m long, into the relief chamber and from there into the Luppmen watercourse. The basin is emptied with a submersible sewage pump after the combined sewer outflow has been removed, or pumped back into the sewer. The basin is cleaned by means of flushing tilts.



Rüb 128 Usterstr

Figure S 2-3: left: 3D drawing of RUB Morgental, incl. Sensor positioning and major flow path during wet weather.

2.4 Rüb 59 ARA(including in-sewer-storage chanel)

The RÜB ARA is a rainwater clarification basin in the shunt. It has a direct storage volume of 540 m3. Until 2020, the inflow to the WWTP was measured by means of a level sensor and regulated to a calculated maximum inflow of 180 l/s-1 via an electric gate valve at the 58_SBW branching structure. Another 172 m3 of impoundment volume is added in the backwater channel to the sewage treatment plant (SK ARA). In August 2020, a combined flow measurement and control system from Stebatec (type TFPNA) was installed at the sewage treatment plant inlet. However, the functionality of the Rüb WWTP has not changed, except that the impoundment point is closer to the WWTP and thus the upstream storage volume has been increased again. In addition, the inflow measurement has become more precise.

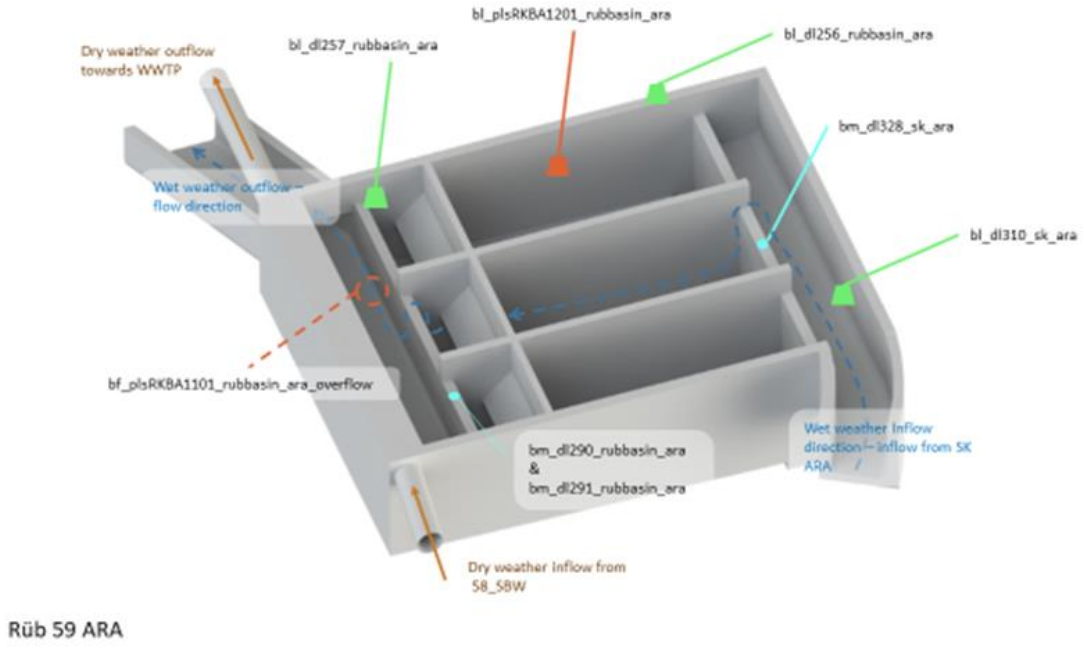


Figure S 2-4 3D drawing of RUB 59 ARA, incl. Sensor positioning and major flow path during wet weather (dotted blue line).

2.5 Rüb PW80 Industry

The flow-through basin with pumping station RB80 Industry has a basin volume of 200 m3. In dry weather, the wastewater is lifted by two pumps and then conveyed to the wastewater treatment plant via a gravity pipe. In the event of rain, the wastewater is discharged into the Kempt. The Tank emptying is coupled with the pumps from the dry weather outflow. The basin is cleaned on demand by a flushing-back-system.

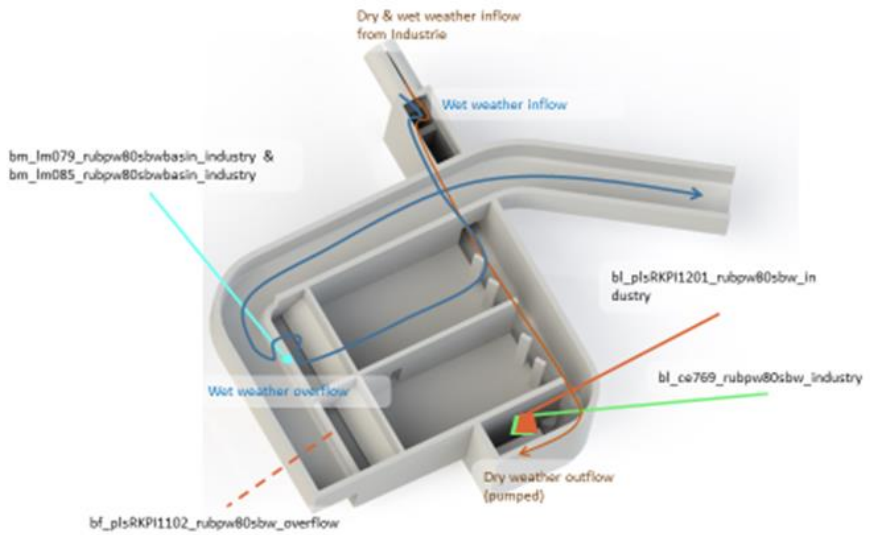


Figure S 2-5: 3D drawing of RUB PW80 Industrie, incl. Sensor positioning and major flow path during wet weather.

2.6 VS600 flow diversion structure

The branching structure VS600 follows after Rüb Morgenthal, conveying the mixed water in the direction of Russikerstrasse and Feldstrasse. However, a built-in plate prevents dry-weather outflow and significant rainfall outflow, with throttling at Rüb Morgenthal (no significant dischargers in between) limited to a maximum of 75 Ls-1. As a result, 100% of the flow goes in the direction of Russikerstrasse, and 0% flows towards Feldstrasse.



Figure S 2-6: Flow diversion structure VS600 with gates (from VGEP 2016).

2.7 VS22 flow diversion structure

The VS22 distribution structure is operated until 2022 by a handwheel(electrified in mid 2022 for maintenance purposes only). In case of heavy rainevents the valve gate couldn't be operated anymore. This is the reason why the valve gate remains nearly fully open and an overflow towards special structure Ra40a practically never happens. The increase in wastewater is moderate from the Rub Morgenthal and further the flow is already throttled there which means it doesn't trigger an overflow with the valve gate open.



Figure S 2-7: Flow diversion structure VS22 with a manually operated gate. The arrow indicates direction of combined sewer overflows (from VGEP 2016).

2.8 Flow control structure Rü40a

Relief system in Fälibach/Luppmen includes a throttle to control the flow, and an open gate position in VS22 is utilized to hold back the flow as needed. Since the open flow gate in upstream structure VS22 never activates an overflow, an overflow can (in this current state) only occur in extremest rain events.



Figure S 2-8: Overflow chamber Rü40a with gates (from VGEP 2016). The left and right pictures show a downstream and upstream view. The numbers 1 and 2 label same points of reference and the arrow indicates downstream flow direction.

2.9 US58

The special storage channel has a volume of 168 m^3. Until mid-2020, a gate valve actuator served as the primary control element to limit flow to the WWTP. However, it now only functions as a maintenance valve gate. A pneumatic TFPNA flow meter (Fa. Stebatec, Brugg CH) has been installed at the inflow from the WWTP to control its inflow further downstream. Since then, the main inflow channel has also been used as an additional storage channel, which is not accounted for in the volume mentioned above.



Figure S 2-9: Storage sewer US58. Until mid-2020, the penstock actuator served as the primary control element to limit the inflow to the WWTP. . The bottom left channel is the main dry weather inflow towards the WWTP, on the right side there is the weir crest visible for the storage channel inlet that runs in parallel towards the RUB ARA (better visible in the right picture).

2.10 Emergency overflow 48b

The emergency relief system is currently inactive. In the rare event that any activity is detected, it is likely to be the result of backwater intrusion into the urban drainage system due to backwater in the nearby Luppmen creek. The WWTP has reported the presence of driftwood in the structures after extreme rainfall or flood events, which supports this assumption.



Figure S 2-10: Emergency overflow 48b, which is only spilling during extremest events.